

Prob & Stats Assignment 03

Name: Yash Raj

Roll-no: 24k-0737

Question #1:-

$$n = 50$$

$$\bar{x} = 1.8 \text{ hours}$$

$$\text{Standard deviation } (s) = \frac{20}{60} = 0.333 \text{ hours}$$

$$\text{Confidence level } 90\%$$

$$Z_{\alpha/2} = 1.645$$

Soln

$$S.E = \frac{s}{\sqrt{n}} = \frac{0.333}{\sqrt{50}} \approx \frac{0.333}{7.071} \approx 0.0471$$

$$M.E = 1.645 \times 0.0471$$

$$[M.E \approx 0.0775]$$

$$e) \text{ Confidence interval: } 1.8 \pm 0.0775$$

$$\text{Lower limit: } 1.8 - 0.0775 = 1.7225$$

$$\text{Upper limit: } 1.8 + 0.0775 = 1.8775$$

$$(1.72, 1.88) \text{ Ans.}$$

Name: Yash Raj

Roll-no: 21k-0727

Question #2

Data: 225, 240, 215, 206, 211, 210, 193, 180,
225, 202.

$n = 10$

C level: 95%

Sol:-

$$\bar{x} = \frac{\sum x}{n} \Rightarrow \frac{2177}{10} = 217.7$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} \quad s \approx 17.64$$

find t-value

$$\left[\begin{array}{l} df = n-1 = 9 \\ t_{0.025, 9} = 2.262 \end{array} \right]$$

$$c) \text{ S.E. } = \frac{s}{\sqrt{n}} \Rightarrow \frac{17.64}{\sqrt{10}} \Rightarrow 5.58$$

$$c) \text{ M.E. } = t \times \text{S.E.} \Rightarrow 2.262 \times 5.58 \approx 12.62$$

2) Confidence interval:

$$217.7 \pm 12.62$$

$$\text{Lower limit: } 217.7 - 12.62 = 205.08$$

$$\text{Upper limit: } 217.7 + 12.62 = 230.32$$

$$= (205.08, 230.32) \quad \text{Ans}$$

Name: Yashraj

Roll-no: 24K-073

Question #3 :-

Avg. Salary is more than 42000
 $n=30$

$$\bar{x} = 43,260$$

$$s = 5230$$

$$\alpha = 0.05$$

Sol:-

$$H_0: \mu \leq 42000$$

$$H_1: \mu > 42000$$

(right-tailed test)

$$Z = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$Z = \frac{43260 - 42000}{5230/\sqrt{30}}$$

$$Z = \frac{1260}{954.81} \approx 1.32$$

$$Z_{0.05} = 1.645$$

$$Z_{\text{calculated}} = 1.32$$

$$Z_{\text{critical}} = 1.645$$

$$1.32 < 1.645$$

H_0 Accepted

Name: Yash Raj

Roll no: 24100777

Question #4:

$$H_0: \mu \leq 24$$

$$H_1: \mu > 24$$

$$n = 50$$

$$\sigma = 28.7$$

$$\alpha = 0.05$$

$$\bar{x} = 29.5$$

Right tailed.

Sol:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{29.5 - 24}{28.7 / \sqrt{50}} \approx 1.36$$

$$[Z = 1.36]$$

P-Value:

$$Z = 1.36 \quad [P = 0.0877]$$

$$P = 0.0877$$

$$\alpha = 0.05$$

$$0.0877 > 0.05$$

∴ fail to reject H_0

[Signature]

Name: Yash Raj
Question #8.

Roll no: 24K-0737

$$\mu = 5.8$$

$$n = 20$$

$$\alpha = 0.05$$

Sol.

$$\bar{x} = 77/20 = 3.85$$

$$s = 2.46$$

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

$$t = \frac{3.85 - 5.8}{2.46/\sqrt{20}}$$

$$t = \frac{-1.95}{2.46/4.472} = \boxed{t = -3.55}$$

$$df = n - 1 = 19$$

$$t_{0.025, 19} = \pm 2.093$$

$$3.55 > 2.093$$

we reject H_0

Name: Yash Raj

Roll-no: 24K0737

Question #C...

Algorithm 1: $n_1 = 12$ $\bar{x}_1 = 85$ $s_1 = 4$

Algorithm 2: $n_2 = 10$ $\bar{x}_2 = 81$ $s_2 = 5$

$\alpha = 0.05$

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$s_p^2 = \frac{401}{20} = 20.05$$

$$s_p = \sqrt{20.05} = 4.48$$

$$[s_p = 4.48] \text{ Ans}$$

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \text{or} \quad \frac{(85 - 81) - 2}{4.48 \sqrt{\frac{1}{12} + \frac{1}{10}}} = \frac{2}{4.48 \sqrt{0.08 + 0.1}}$$

$$t = \frac{2}{4.48 \sqrt{0.1833}} \quad t = \frac{2}{4.48 (0.428)} \quad \boxed{t = \frac{2}{1.918}}$$

$$\boxed{t \approx 1.04}$$

$$df = n_1 + n_2 - 2 = 20$$

$$t_{0.05, 20} = 1.728$$

$$1.04 < 1.728$$

H_0 accepted

